Kent Pearce **MATH 1351 Spring 2004**

Homework Set HW13 due 4/28/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)
This is problem 6 Section 5.4 page 308.
Evaluate the definite integral

$\int_{-1}^{1}(x^3+bx^2)\,dx =$ ______

2.(1 pt)
This is problem 8 Section 5.4 page 308.
Evaluate the definite integral

$\int_{-2}^{-1}\frac{p}{x^2}\,dx =$ ______

3.(1 pt)
This is problem 12 Section 5.4 page 308.
Evaluate the definite integral

$\int_{0}^{1}(7x^8+\sqrt{\pi})\,dx =$ ______

4.(1 pt)
This is problem 16 Section 5.4 page 308.
Evaluate the definite integral

$\int_{e^{-2}}^{e}\frac{dx}{x} =$ ______

5.(1 pt)
This is problem 18 Section 5.4 page 308.
Evaluate the definite integral

$\int_{0}^{1}\sqrt{t}(t-\sqrt{t})\,dt =$ ______

6.(1 pt)
This is problem 20 Section 5.4 page 308.
Evaluate the definite integral

$\int_{1}^{4}\frac{x^2+x-1}{\sqrt{x}}\,dx =$ ______

7.(1 pt)
This is problem 24 Section 5.4 page 308.
Evaluate the definite integral

$\int_{0}^{\pi/4}(\sec^2(x)-\tan^2(x))\,dx =$ ______

8.(1 pt)
This is problem 26 Section 5.4 page 308.
Evaluate the definite integral

$\int_{1}^{2}\frac{x^3+1}{x}\,dx =$ ______

9.(1 pt)
This is problem 32 Section 5.4 page 308.
Find the area of the region under the curve over the prescribed interval

$y=\sqrt{t}$ on $[0,1]$

AREA: ______

10.(1 pt)
This is problem 34 Section 5.4 page 308.
Find the area of the region under the curve over the prescribed interval

$y=\sin(x)+\cos(x)$ on $[0,\pi/2]$

AREA: ______

11.(1 pt)
This is problem 38 Section 5.4 page 308.
Find the area of the region under the curve over the prescribed interval

$y=\frac{2}{1+t^2}$ on $[0,1]$

AREA: ______

12.(1 pt)
This is problem 40 Section 5.4 page 308.
Find the derivative of the given function

$F(x)=\int_{-2}^{x}(t+1)\sqrt[3]{t}\,dt$

$F'(x) =$ ______

13.(1 pt)
This is problem 42 Section 5.4 page 308.
Find the derivative of the given function

$F(t)=\int_{t}^{2}\frac{e^x}{x}\,dx$

$F'(x) =$ ______

14.(1 pt)
This is problem 1 Section 5.5 page 315.
Evaluate the definite integrals

(a) $\int_{0}^{4}(2t+4)\,dt =$ ______

(b) $\int_{0}^{4}(2t+4)^{-1/2}\,dt =$ ______

15.(1 pt)
This is problem 6 Section 5.5 page 315.
Evaluate the indefinite integrals using a substitution

(a) $\int x(3x^2-5)\,dx =$ ______ $+C$

(b) $\int x(3x^2-5)^{50}\,dx =$ ______ $+C$

16.(1 pt)
This is problem 8 Section 5.5 page 315.
Evaluate the indefinite integrals

(a) $\int\frac{dx}{\sqrt{1-x^2}} =$ ______ $+C$

(b) $\int\frac{x\,dx}{\sqrt{1-x^2}} =$ ______ $+C$

17.(1 pt)
This is problem 10 Section 5.5 page 315.
Use a substitution to evaluate the indefinite integral

$\int\sqrt{3t-5}\,dt =$ ______ $+C$

18.(1 pt)
This is problem 14 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$\int \csc^2(5t)\,dt =$ ______________ $+C$

19.(1 pt)
This is problem 18 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$\int \frac{(6x-9)}{(x^2-3x+5)^3}\,dx =$ ______________ $+C$

20.(1 pt)
This is problem 20 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$\int \sin^3(t)\cos(t)\,dt =$ ______________ $+C$

21.(1 pt)
This is problem 24 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$\int \frac{4x\,dx}{2x+1} =$ ______________ $+C$

22.(1 pt)
This is problem 30 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$\int \frac{\ln(x+1)}{(x+1)}\,dx =$ ______________ $+C$

23.(1 pt)
This is problem 34 Section 5.5 page 315.

Use a substitution to evaluate the indefinite integral

$\int \frac{e^{\sqrt{t}}}{\sqrt{t}(e^{\sqrt{t}}+1)}\,dt =$ ______________ $+C$

24.(1 pt)
This is problem 36 Section 5.5 page 315.

Evaluate the definite integral

$\int_1^4 \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx =$ ______________ $+C$

25.(1 pt)
This is problem 40 Section 5.5 page 315.

Evaluate the definite integral

$\int_0^2 x\sqrt{2x+1}\,dx =$ ______________ $+C$

26.(1 pt)
This is problem 21 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$\frac{dy}{dx} = 3xy$

NOTE: if your answer is expressed in implicit form as $y = Cf(x)$ where C is a constant, then your answer would be $f(x)$, e.g., you might have $C\sin(x^2)$ and in this case you would only enter $\sin(x^2)$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$y =\ C$ ______________
where C is a constant.

27.(1 pt)
This is problem 22 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$\frac{dy}{dx} = \sqrt{\frac{y}{x}}$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y)$, e.g., you might have $\sin(x^2-y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2-y^2)$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$f(x,y) =$ ______________
where C is a constant.

28.(1 pt)
This is problem 24 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$\frac{dy}{dx} = (y-4)^2$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y)-C$, e.g., you might have $\sin(x^2-y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2-y^2)$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$f(x,y) =$ ______________
where C is a constant.

29.(1 pt)
This is problem 26 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$\frac{dy}{dx} = \frac{y}{\sqrt{1-2x^2}}$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y)$, e.g., you might have $\sin(x^2-y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2-y^2)$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$f(x,y) =$ ______________
where C is a constant.

(Note: the notation for absolute value of a number y is *abs(y)*)

30.(1 pt)
This is problem 30 Section 5.6 page 326.

Find the general solution solution of the separable differential equation.

$\frac{dy}{dx} = e^{y-x}$

NOTE: if your answer is expressed in implicit form as $f(x,y) = C$ where C is a constant, then your answer would be $f(x,y)$, e.g., you might have $\sin(x^2-y^2) = C$ and in this case you would only enter the left hand side $\sin(x^2-y^2)$. YOU MUST US CAPITAL C FOR THE CONSTANT.

$f(x,y) =$ ______________
where C is a constant.